

A Reliable Networking Platform For Reinforcement Learning Researchers

113-1 (Fall 2024), Reinforcement Learning
Final Project

組別: RL4Life

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2024/12/18

Background

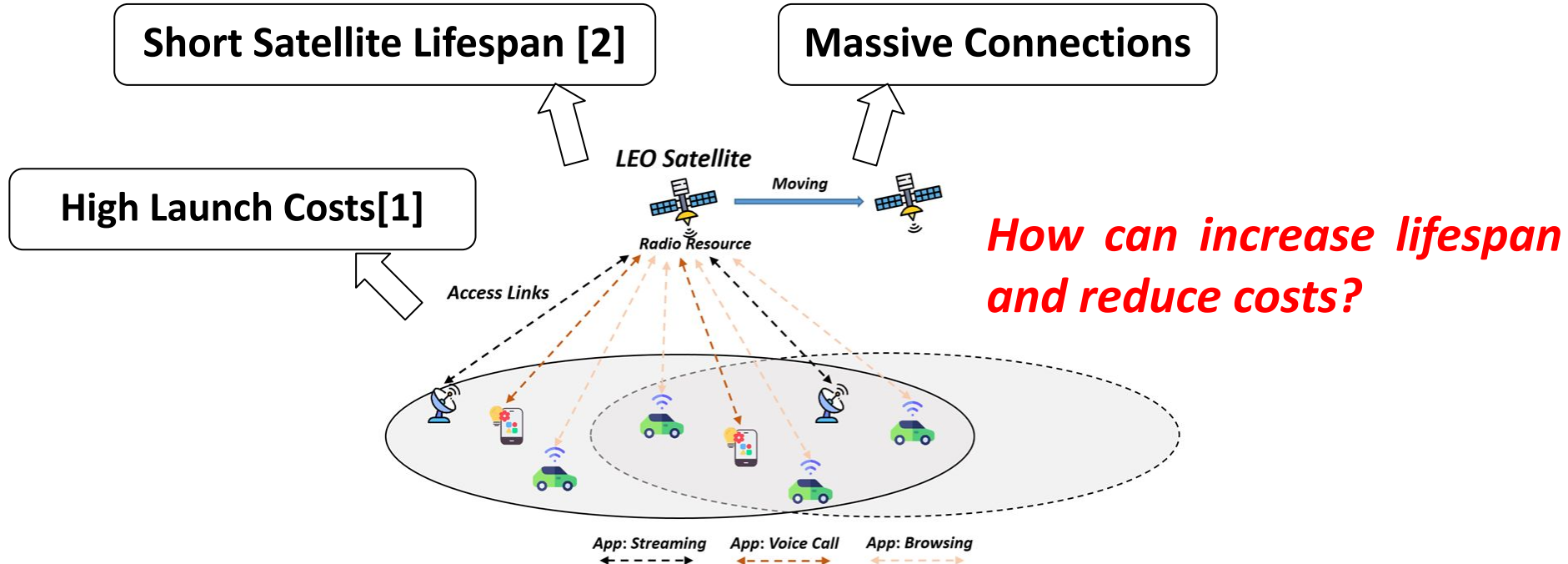


Fig. 1. A LEO Satellite Providing Network Services

(Background) Nowadays, many researches discussed the Intelligent RL-based Power Control

[1] <https://spacenews.com/spacex-and-the-categorical-imperative-to-achieve-low-launch-cost/>

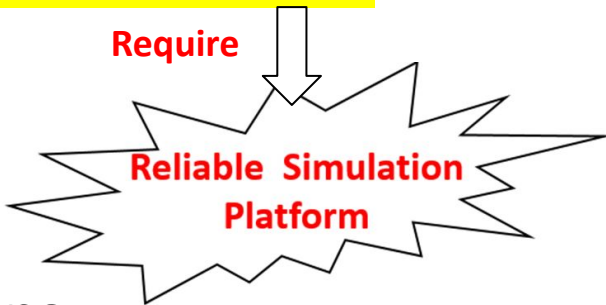
[2] <https://www.space.com/spacex-starlink-satellites.html>

Motivation

Challenges of Operator:

(Limit Computational Resource): Can not put the model on board → **No Online Learning**

★ “Table Look Up” will be the best solution.

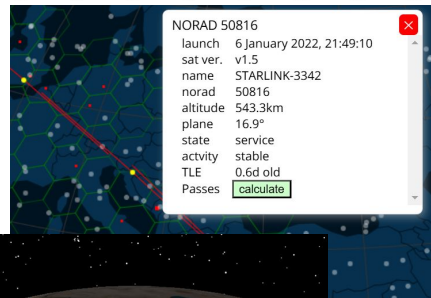
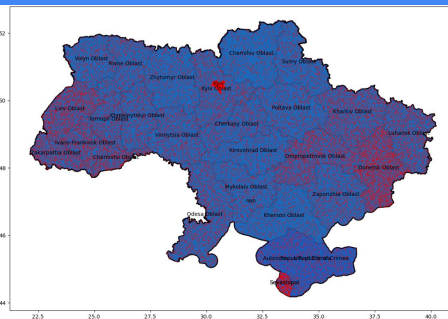
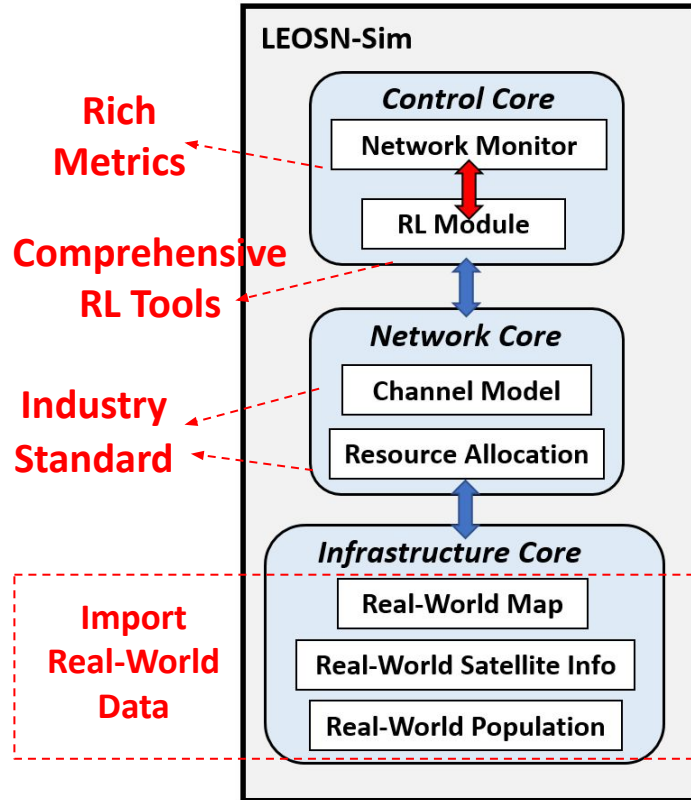


Current Research Limitations:

- Vary user-defined environments → **No fair algorithm comparisons**
- Vary user-defined Settings → **Hard to guarantee practicality**

(Motivation): There is no platforms like OpenAI Gym for addressing network issues.

Contributions

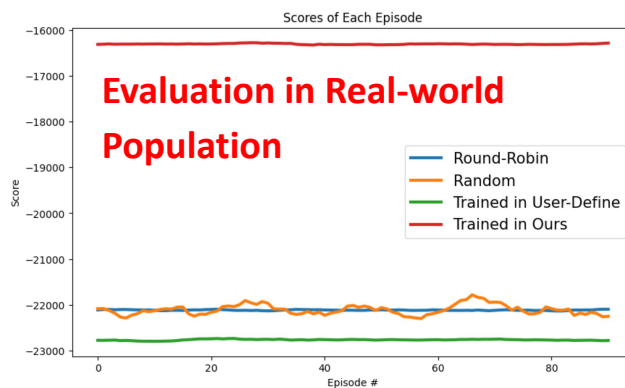
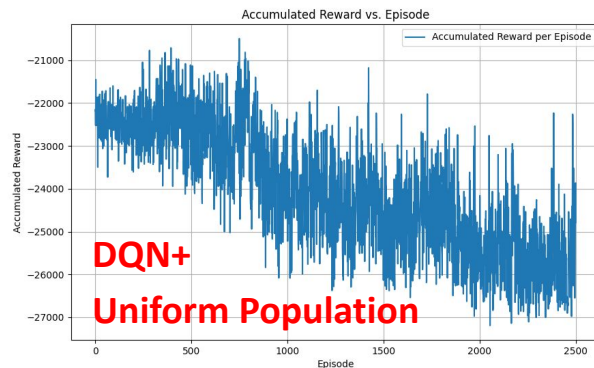
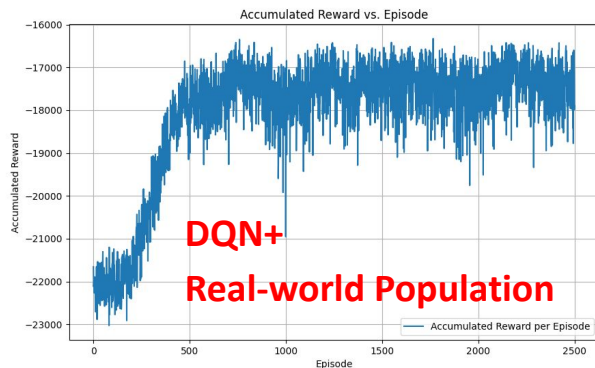


1. ☐ select map() & satellite()
()
 2. load.RL.Module()
 3. model.fit()
 4. model.test()
-

We want a powerful simulator for networking-related RL research, similar in spirit to OpenAI Gym

Evaluation (Effect of Population)

- We evaluate the effect of “Population”, to investigate how “Population” affects the learning process and evaluation.



[Insight] Given the same training settings, we believe the "real-world population" in the infrastructure core plays a crucial role and can significantly influence learning performance. (More Information can be found in report)